1 Numerator $\approx 1.907878...$, Denominator $= 1.025$ **[M1]**

1.861344784 **[A1]**

[Total Marks: 2]

2 (a) 3.068822... **[B2]**

If 0 scored, allow [B1] for seeing 15.36 or 2.485... or 0.81

(b) 3.069 [B1] (strict FT from their (a) if given to at least 4 d.p.)

[Total Marks: 3]

3 (a) Numerator $\sqrt{22.01} = 4.69148\ldots$ or Denominator $12.167 - 1.4 = 10.767$ seen **[M1]**

0.435727839 **[A1]**

(b) 0.436 **[B1]**

[Total Marks: 3]

4 (a) M1 for $\sqrt{158.79}$ or $12.60119\ldots$ or 4.48 seen

2.812765825 (accept to at least 4 decimal places) **[A1]**

(b) 2.81 **[B1ft]**

(c) 3.75×10^8 **[B1]**

[Total Marks: 4]

5 (a) 64.79 or 14.861 or ≈ 3.855 seen **[M1]**

16.80675747 (allow up to 2 missing trailing digits depending on calculator model) **[A1]**

(b) 16.8 **[B1]**

(c) 6 hours 27 minutes **[B1]**

[Total Marks: 4]

6 (a) 4.423... or $\dfrac{15.98...}{3.612}$ **[M1]**

4.42 or 4.424 **[A1]**

(b) $1.4862... - 0.8191...$ or 0.6671... **[M1]**

0.667 **[A1]**

[Total Marks: 4]

7 (a) 4.8 **[M1]**

4.80 **[A1]**

(b) $2 + \frac{18}{60}$ or $\frac{138}{60}$ **[M1]**

2.3 **[A1]**

[Total Marks: 4]

8 (a) $150.0625 + 4.4412...$ or $154.5037...$ **[M1]**

154.50 **[A1]**

(b) $\frac{1.3639...}{0.9510...}$ or $1.434...$ **[M1]**

1.43 **[A1]**

[Total Marks: 4]

9 (a) 8.40 **[B1]**

(b) 5 hours 9 minutes **[B1]**

(c) $280 \div 85$ **[M1]**

3.294... **[A1]**

3 hours 18 minutes **[A1]**

[Total Marks: 5]

10 (a) (i) Numerator 4.94 or denominator 1.14... seen **[M1]**

4.332666... **[A1]**

(ii) 4.33 **[B1]**

(b) 0.35×60 **[M1]**

5 h 21 min **[A1]**

[Total Marks: 5]

11 (a) $\pi \times 4.2^2$ or $55.417...$ **[M1]**

55 **[A1]**

(b) $23 \div 60$ or $0.3833...$ seen **[M1]**

Time $= 4 + \frac{23}{60}$ or $4.3833...$ hours **[M1]**

Distance $= 78 \times 4.3833...$ **[M1]**

341.9 **[A1]**

[Total Marks: 6]

12 **(a)** Time $= 3 + \frac{40}{60}$ or $\frac{11}{3}$ hours **[M1]**

Cost $= 45 + 32.50 \times \frac{11}{3}$ **[M1]**

164.17 **[A1]**

(b) $230.25 - 45 = 185.25$ **[M1]**

$\frac{185.25}{32.50} = 5.7$ hours **[M1]**

5 hours 42 minutes **[A1]**

[Total Marks: 6]

13 **(a)** M1 for $\frac{4}{3} \times \pi \times 4.85^3$

477.8 **[A1]**

(b) M1 for $\frac{1}{3} \times \pi \times 1.2^2 \times 3.5$ (implies $5.2778\ldots$)

M1 for $\frac{\text{their } 477.8}{\text{their } 5.2778\ldots}$

A1 for $90.5\ldots$

90 **[B1]**

(c) 0.584 (accept 0.582 to 0.585) **[B1ft]**

[Total Marks: 7]

14 **(a)** Time $= 24\frac{40}{60}$ or 24.666... or $\frac{74}{3}$ **[M1]**

$\frac{74}{3} \times 18.60 = 458.8$ **[M1]**

458.80 (must have trailing zero) **[A1]**

(b) 479.45 **[B1]**

(c) 3.4×150 or 510 minutes **[M1]**

$510 \div 60$ or 8.5 **[M1]**

8 hours 30 minutes **[A1]**

[Total Marks: 7]

15 **(a)** $\frac{4}{3}\pi(1.4)^3$ **[M1]**

11.49404... **[A1]**

(b) their 11.494... $\div 0.035$ **[M1]**

328.4... or 328 (minutes) **[A1]**

their 328.4... $\div 60$ (converting to hours) **[M1]**

5 hours 28 minutes **[A1]**

[Total Marks: 6]

16 **(a)** 14.4×10^3 or 14400 **[M1]**

1.44×10^4 **[A1]**

(b) $\dfrac{1}{2.5066...} \times 0.7788...$ or $0.398942... \times 0.778800...$ or $0.310696...$ **[M1]**

0.3107 **[A1]**

(c) $1.5 = 2\pi\sqrt{\dfrac{l}{9.81}} \implies \dfrac{1.5}{2\pi} = \sqrt{\dfrac{l}{9.81}} \implies 0.2387... = \sqrt{\dfrac{l}{9.81}}$ **[M1]**

$l = 9.81 \times (0.2387...)^2 \implies \mathrm{l} = 0.55909...$ **[M1]**

0.56 **[A1]**

[Total Marks: 6]

17 **(a)** Numerator $= 340000 + 27000 = 367000$ or 3.67×10^5 **[M1]**

$3.67 \times 10^5 \div (8.1 \times 10^{-3}) = 45308641.98...$ **[M1]**

4.53×10^7 or 4.531×10^7 **[A1]**

(b) Upper bound for $r = 5.25$ and Upper bound for $h = 12.45$ **[M1]**

$V = \dfrac{1}{3} \times \pi \times 5.25^2 \times 12.45$ **[M1]**

359.34... or 359 **[A1]**

[Total Marks: 6]

18 **(a)** M1 for $2\pi\sqrt{\dfrac{0.85}{9.81}}$

1.850 **[A1]**

(b) M1 for $\left(\dfrac{T}{2\pi}\right)^2 = \dfrac{l}{g}$ or $\dfrac{T^2}{4\pi^2} = \dfrac{l}{g}$

$\dfrac{gT^2}{4\pi^2}$ **[A1]**

(c) M1 for their $l = \dfrac{9.81 \times 2.5^2}{4\pi^2}$

A1 for $l = 1.5532\ldots$ metres

155.3 **[A1]**

[Total Marks: 7]

19 **(a)** $\dfrac{1}{3} \times \pi \times 3.2^2 \times (3 \times 8.45 - 3.2)$ **[M1]**

237.5 **[A1]**

(b) $\dfrac{3V}{\pi h^2} = 3R - h$ **[M1]**

$\dfrac{3V}{\pi h^2} + h = 3R$ **[M1]**

$R = \dfrac{V}{\pi h^2} + \dfrac{h}{3}$ or $R = \dfrac{3V + \pi h^3}{3\pi h^2}$ **[A1]**

(c) $\dfrac{500}{\pi \times 5.6^2} + \dfrac{5.6}{3}$ **[M1]**

6.94 **[A1]**

[Total Marks: 7]

20 (a) $\dfrac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.80 \times 10^6}$ or $5.8558... \times 10^7$ **[M1]**

$\sqrt{5.8558... \times 10^7}$ or 7652.36... **[M1]**

7650 **[A1]**

(b) Total distance $= 2 \times \pi \times 6.80 \times 10^6$ or 42725660... **[M1]**

Time in seconds $= \dfrac{42725660...}{7652.36...} = 5583.32...$ **[M1]**

$5583 \div 3600 = 1.55...$ (implies 1 hour) and remainder 1983 seconds **[M1]**

1 hour 33 minutes 3 seconds **[A1]**

[Total Marks: 7]

21 (a) $\dfrac{1}{3} \times \pi \times 2.31^2 \times (3 \times 8.74 - 2.31)$ **[M1]**

133.607... **[A1]**

133.6 **[A1]**

(b) Rounding initial values: $R = 8.7$ and $h = 2.3$ **[B1]**

$\dfrac{1}{3} \times \pi \times 2.3^2 \times (3 \times 8.7 - 2.3)$ or 131.84... **[M1]**

Difference $= 133.607... - 131.844... = 1.76...$ **[M1]**

1.8 **[A1]**

[Total Marks: 7]